query, even in a distributed setup. OpenSearch started out as a fork of Elasticsearch, sponsored by Amazon, and reached general availability (http://mng.bz/a12j) with its 1.0 version in July 2021. As time goes by, the ES/OS code bases are expected to diverge, but from the get-go, the major difference was the license under which each project was distributed. However, ES and OS have the fact that they are logs-first backends in common (optimized for storage and retrieval of logs). As a reminder, in chapter 4, we used OpenSearch as a destination for the Fluent Bit agent.

I won't cover ES/OS in detail in this book, as there are plenty of resources available on the topic, so if you want to dive deep, I'd recommend reading the excellent second edition of the Manning book *Elasticsearch in Action* by Madhusudhan Konda (2023; https://www.manning.com/books/elasticsearch-in-action-second-edition).

We will get back to the ES/OS topic in the context of traces later in this chapter, since this is a signal type ES/OS have decided to cover nowadays as well (support for metrics is also available; however, it is not as full blown as for the other two types).

- *Grafana Loki (https://grafana.com/oss/loki/)*—This is a horizontally scalable, multitenant log aggregation system, released under the AGPLv3 license, and its design was inspired by Prometheus. It does not index the contents of the logs themselves (as ES/OS do) but rather a set of labels for the ingested log streams. You have seen Loki in action already in chapter 2, and it's certainly something to take a closer look at; currently, few managed offerings based on it exist.
- *ZincObserve (https://github.com/zinclabs/zincobserve)*—This is a search engine you can use for indexing and looking up logs, with a focus on reducing log storage costs.

Now, let's move on to commercial offerings in the space.

5.2.3 Commercial offerings for log backends

The commercial offering space for log backends is well set up, and you will find some common features across of all of these, such as paid ingestion, strong multitenancy support, and a wide range of built-in integrations with other signal sources, from apps to cloud provider offerings. The offerings include but are not limited to the following:

- *Splunk (https://www.splunk.com/)*—The established logs incumbent, with a range of enterprise features and strong OpenTelemetry commitment
- *Instana (https://www.instana.com/)*—Acquired by IBM and provides many interesting features and integrations
- *SolarWinds (https://www.solarwinds.com/)*—Acquired formerly independent SaaS offerings Loggly and Papertrail as well as the monitoring offering Pingdom
- *Logz.io (https://logz.io/)*—A start-up with an amazing range of offerings in the cloud-native domain (using OpenSearch and Jaeger)